SHORT-PAPER

A Large-Scale Study of Reranker Relevance Feedback at Inference

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A Large-Scale Study of Reranker Relevance Feedback at Inference

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Abstract

Neural IR systems often employ a retrieve-and-rerank framework: a bi-encoder retrieves a fixed number of candidates (*e.g.*, K=100), which a cross-encoder then reranks. Recent studies have indicated that relevance feedback from the reranker *at inference time* can improve the recall of the retriever. The approach works by updating the retriever's query representations via a distillation process that aligns it with the reranker's predictions. While a powerful idea, the arguably narrow scope of past studies focusing on a small number of specific domains such as english question answering and entity retrieval has left a gap in our understanding of how well it generalizes. In this paper, we study inference-time reranker relevance feedback extensively across multiple retrieval domains, languages, and modalities, while also investigating aspects such as the performance and latency implications of the number of distillation updates and feedback candidates.

CCS Concepts

• **Information systems** → **Query representation**; **Relevance assessment**.

Keywords

Relevance Feedback; Query Representation; Distillation; Retrieve-and-Rerank

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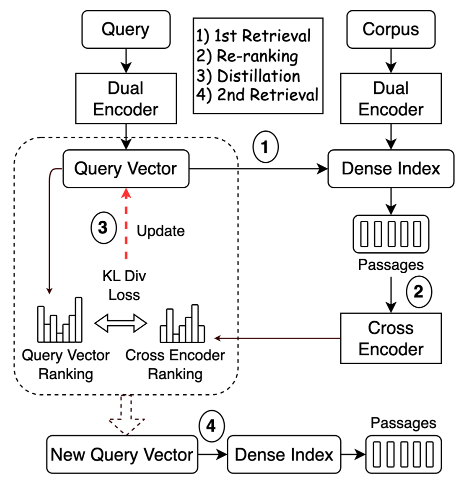

Figure 1: Illustration of ReFIT: Reranker Relevance Feedback at Inference-Time. The distillation process (step 3) updates the query vector in a traditional retrieve-and-rerank framework (steps 1 and 2), improving recall when used for a second retrieval step (step 4).

1 Introduction

Retrieve-and-rerank (R&R) [6] in neural information retrieval (IR) often employs a *dual-encoder* retriever to retrieve an initial set of candidates in response to a user query [9] and a *cross-encoder* reranker to rerank them in order of their relevance [15]. While cross-encoders typically offer superior ranking performance, their application is crucially limited to reranking only the top K candidates retrieved by the dual-encoder. Reranking thus does not affect retrieval metrics such as Recall@K, that evaluate the presence of relevant candidates in the top K results. Achieving a high recall, however, can be key in many use cases, especially when retrieved results are used in downstream knowledge-intensive tasks [16].

Some recent studies [22, 27] have proposed improving the recall of retrievers by using relevance feedback from a reranker. The process works by first updating the retriever's query vector via knowledge distillation from the reranker at inference time, which brings the score distributions of the two models over the top K retrieved passages closer. The new query vector is then used to retrieve documents a second time. This approach shares similarities with pseudo-relevance feedback (PRF) [13, 20] in classical IR, which has been used with sparse approaches via query expansion [1, 25]. While using rerankers for relevance feedback can offer distinct advantages over prior architecture- and language-specific dense PRF methods [2, 28], their efficacy has empirically been tested only on a few narrow use cases [22]. This leaves room for further examination of the utility of this promising approach. Moreover, existing work [5, 7, 17, 24] primarily integrates reranker feedback

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during training of the dual-encoder retriever. In contrast, distillation at inference time updates only the query representation to replicate the reranker scores for a given test instance. This design keeps retriever parameters unchanged, allowing seamless integration into any neural retrieval-and-reranking framework. Conversely, training-time distillation necessitates retraining the bi-encoder for new languages or modalities.

Given the promise of inference-time reranker relevance feedback, we carry out a large-scale asessment of this approach, involving multi-domain, multilingual and multimodal evaluation. With our implementation, termed ReFIT (§2), we demonstrate that reranker relevance feedback generalizes well to a variety of R&R use cases (§3.3). We also present additional analysis of ReFIT, including a comparison with training-time reranker distillation (§3.4.1) and a latency-versus-performance tradeoff of different aspects of its underlying distillation mechanism (§3.4.3). We also compare ReFIT with TouR [22] (in §3.4.4), another approach for test-time optimization of query representation using reranker scores, that was previously only evaluated on English phrase and passage retrieval.

2 ReFIT: Reranker Relevance Feedback

In this section, we describe how ReFIT exploits relevance feedback from a high-accuracy reranker to update the query representations of a retriever during inference (Figure 1). Given a query q and a passage p (or an equivalent unit of retrieval from a non-text modality), let E_q and E_p be their vector representations computed by a bi-encoder retriever θ. A common measure of similarity between E_q and E_p, a.k.a. the *retriever score*, is their dot product: $sim(q, p) = E_q^T E_p$. Let $R(q, p)$ be the corresponding *reranker score*.

ReFIT performs a lightweight inference-time distillation of $R(q, p)$ scores into E_q. More specifically, given retrieved passages $P = \{p_i\}_{i=1}^K$, the reranker ϕ first produces a set of scores $R(q, P) = \{R(q, p_i)\}_{i=1}^K$. A reranker scoring distribution is then computed over the elements of P:

$$D_\phi(q, P) = \text{softmax}([R(q, p_1), ..., R(q, p_K)]) \quad (1)$$

A similar distribution is also computed over the retriever scores:

$$D_\theta(q, P) = \text{softmax}([E_q^T E_{p_1}, ..., E_q^T E_{p_K}]) \quad (2)$$

ReFIT minimizes the following KL-divergence loss during training:

$$\mathcal{L} = D_{KL}(D_\phi(q, P) || D_\theta(q, P)) \quad (3)$$

and updates the query vector via gradient descent. The update process is repeated n times, yielding a new query vector $E_q^{(n)}$, where n is a hyper-parameter. A step-by-step description of the process is given in Algorithm 1. The final step in ReFIT is to retrieve a second time from the passage index with the new vector $E_q^{(n)}$, with an aim of improving recall over the original vector E_q while retaining the reranking accuracy of the distilled reranker.

Algorithm 1 ReFIT

Input: Query q and its representation E_q, retrieved passages $P = [p_1, ..., p_K]$ and their representations $[E_{p_1}, ..., E_{p_K}]$

Output: Updated query representation $E_q^{(n)}$

```
Initialize query vector E_q^(0) = E_q
Compute reranker scores D_φ(q, P) (Eq. 1)
for i in 0 to n − 1 do
    Compute retriever scores D_θ(q, P) with E_q^(i) (Eq. 2)
    Compute loss ℒ (Eq. 3)
    Update E_q^(i+1) = E_q^(i) − α ∂/∂E_q^(i) ℒ
end for
```

3 Experiments

3.1 Setup

For a comprehensive assessment of inference-time reranker feedback, we evaluate ReFIT in multi-domain, multilingual, and multimodal settings.

Models: We use Contriever [8] as the text retriever, while the English[1] and multilingual[2] rerankers are based on sentence transformers. In multimodal evaluation, we use BLIP [11], a state-of-the-art vision-language model comprising two unimodal encoders serving as the bi-encoder retriever, and an image-grounded text encoder functioning as the cross-encoder reranker.

Distillation: We apply min-max normalization separately to the query vector's scores $Q_q^T \hat{P}$ and reranker scores $R(q, P)$ to align their distributions. Since the reranker produces peaky distributions, we use a temperature $T = 2$ (tuned) in the softmax $D_{CE}(q, P)$. After tuning on the MS MARCO dev set, we set $n = 100$ updates with a learning rate $\alpha = 0.005$.

Datasets: We use BEIR [23] for English multi-domain evaluation, which includes in-domain MS MARCO [14] as well as out-of-domain test data from various scientific, biomedical, financial, and Wikipedia-based retrieval datasets. For multilingual evaluation, we use two Wikipedia-based benchmarks Mr.TyDi [3] and MKQA [12]; the former involves retrieval of passages that are in the same language as the query (across 11 languages) while the latter asks to retrieve from the English Wikipedia given queries in 26 different languages. For multimodal evaluation, we use MSRVTT [26], which involves text-to-video retrieval.

3.2 Latency Considerations

ReFIT introduces additional overhead to the R&R framework due to distillation and a second retrieval step. Distillation latency scales linearly with the number of updates n (Algorithm 1). Table 1 compares the inference latency of ReFIT and standard R&R for text, assuming n=100 updates during distillation. The lightweight distillation process takes only 30ms on a CPU. The additional steps increase inference latency for the full pipeline by approximately 4.4% on a CPU and 17.5% on a GPU[3]. In that same amount of time, vanilla R&R can process K=125 passages on the GPU (Table 1), to potentially increase Recall@100. Given this observation, and for a fair comparison, we compare ReFIT with a *Rerank* baseline that reranks K=125 passages instead, while evaluating both on Recall@100.

3.3 Main Results

In Table 2 we show the results of evaluating ReFIT in English multi-domain, multilingual and multimodal settings. While reranking

[1] cross-encoder/ms-marco-MiniLM-L-6-v2
[2] cross-encoder/mmarco-mMiniLMv2-L12-H384-v1
[3] 24-core AMD EPYC 7352 CPU and 80GB A100 GPU.

Step (Device)	Retrieve & Rerank		ReFIT (K=100)
	K=100	K=125	
1st Retrieval (CPU)	40	40	40
Rerank (CPU/GPU)	1540/360	1925/450	1540/360
Distillation (CPU)	-	-	30
2nd Retrieval (CPU)	-	-	40
Total (CPU/GPU)	1580/400	1965/490	1650/470

Table 1: Comparison of inference times (in milliseconds) for different approaches, utilizing both CPU-only and GPU configurations, when reranking K passages.

with K=125 expectedly improves Recall@100 over the retriever, ReFIT consistently outperforms both. These results underscore the advantage of ReFIT's architecture independence and compatibility with arbitrary R&R frameworks. While ReFIT primarily aims to improve retriever recall, the ranking performance after the second retrieval step matches that of the reranker, thereby needing no additional reranking after (discussed in §3.4.2).

	Multi-Domain		Multilingual		Multimodal
	MS Marco	BEIR	Mr.TyDi	MKQA	MSRVTT
Retriever	89.1	66.8	87.0	65.6	92.1
Reranker	89.9	67.6	88.1	66.4	92.3
ReFIT	**90.5***	**69.0***	**89.7***	**68.2***	**92.9**$^+$

Table 2: Recall@100 for evaluation in multi-domain, multilingual, and multimodal settings. * and $^+$ correspond to statistically significant improvements at p<0.05 and 0.1, respectively, as per a paired t-test.

3.4 Analysis

3.4.1 ReFIT vs Training-time Distillation. We compare ReFIT's inference-time relevance feedback with RocketQAv2 [19], a training-time distillation method. RocketQAv2 builds on top of the RocketQAv1 [18] retriever by leveraging reranker scores for distillation during training. As shown in Table 3, RocketQAv2 outperforms RocketQAv1 on MS MARCO–the dataset used for its distillation–but underperforms on out-of-domain test sets from BEIR. This outcome is not surprising, as training-time distillation restricts the retriever to seeing the reranker's relevance labels only in the source domain(s), limiting generalization to unseen domains. In contrast, ReFIT benefits from target-domain pseudolabels provided by the reranker at inference, enhancing out-of-domain generalization.

3.4.2 Recall Improves, but How Good is the Ranking? In §3.3, we demonstrated that ReFIT improves Recall@100 over existing retrieval and reranking baselines. In many applications including web search, the ranking of retrieved results matters too. An essential question for our method, therefore, is whether relevance feedback contributes to an accurate ranking of the retrieved results; a failure to do so would necessitate an additional, computationally expensive reranking step. To assess the ranking performance of ReFIT, we conduct evaluations on the BEIR and Mr. TyDi. The results, presented in Table 4, report nDCG@100 scores, as this metric comprehensively captures variations in ranking performance across a

	Distillation	MS Marco	BEIR
RocketQAv2	Training-time	88.7	61.7
RocketQAv1	None	88.4	63.1
Reranker (K=125)	None	89.4	64.4
ReFIT (K=100)	Inference-time	**90.0**	**65.9**

Table 3: Recall@100 numbers for comparison against RocketQAv2, a training-time distillation baseline. Rerank and ReFIT use RocketQAv1 as the retriever.

	BEIR	Mr.TyDi
Retriever	48.1	48.0
Reranker (K=100)	51.3	63.7
Reranker (K=125)	51.6	63.9
ReFIT (K=100)	**52.5**	**64.1**

Table 4: nDCG@100 numbers for ranking performance.

larger set of retrieval outputs. While *Reranker* expectedly yields big gains over the retriever, ReFIT is able to improve over *Reranker* on ranking performance. As a result, it eliminates the need for any additional reranking following relevance feedback. These findings indicate that ReFIT is well-suited for applications requiring both high retrieval recall and accurate ranking.

3.4.3 Latency vs Performance Tradeoff. Here we investigate the effect of various aspects of ReFIT's distillation process on its performance as well as the additional latency they introduce. We consider number of distillation updates n, number of passages reranked and subsequently used for distillation K, along with extending the relevance feedback to multiple iterations.

Distillation Updates: The latency of distillation is linear in the number of updates n, with the additional retrieval step taking constant time. In Figure 2 (left), we see that for a mere 4.4% increase in latency (n=100), ReFIT considerably boosts performance over computationally expensive reranking of K=125 candidates (24.3% increase in latency). Under stricter latency constraints, ReFIT can be made faster by simply lowering the number of updates, while still surpassing the *Rerank* baseline.

Passages Reranked: We investigate how ReFIT performs as we vary K, the number of passages reranked and subsequently used for distillation. Smaller values of K correspond to a faster R&R pipeline, but it comes at the expense of the target reranker distribution containing less information. In Figure 2 (middle), higher values of K expectedly lead to higher recall in general, while we observe performance improvements over the *Rerank* baseline even with considerably smaller K values.

Relevance Feedback Iterations: Finally we examine if further gains are possible from more iterations of relevance feedback, i.e., running the following steps in a loop: (1) rerank the retrieved documents from the previous iteration, (2) update the query vector via distillation from the reranker distribution, and (3) retrieve again. In Figure 2 (right), we see that recall does go up with more rounds of relevance feedback; the biggest gain, however, comes in the first round, while later rounds give diminishing returns along with a considerable increase in latency from needing to perform the computationally expensive reranking at every iteration.

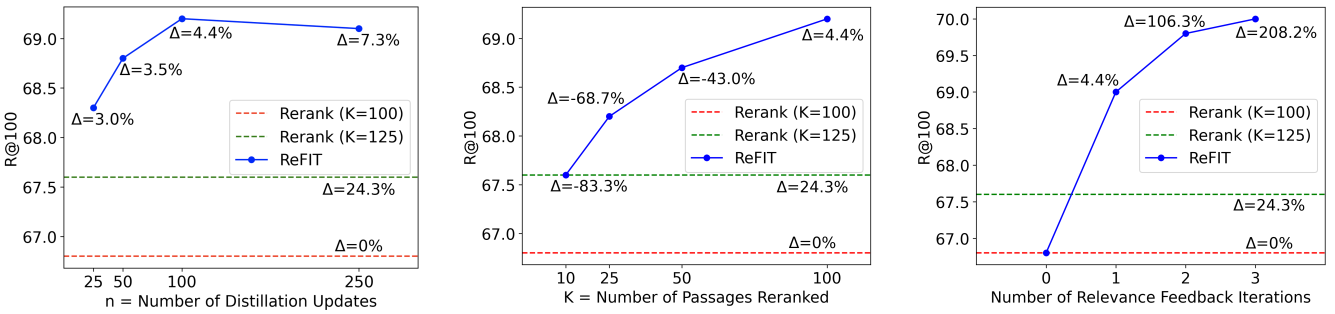

Figure 2: Variation of ReFIT performance with distillation updates n (left), reranked passages K used for distillation supervision (middle) and relevance feedback iterations (right). Δ corresponds to change in latency with respect to the standard R&R framework with K=100 (CPU-only configuration).

	NQ	EntityQ	BEIR	Mr.TyDi
Retriever	86.1	70.1	66.8	87.0
Reranker (K=125)	86.8	71.2	67.6	88.1
TouR$_{\text{hard}}$ (K=100)	87.0	71.9	68.4	88.7
TouR$_{\text{soft}}$ (K=100)	87.2	72.5	68.1	88.1
ReFIT (K=100)	**87.6**	**72.6**	**69.0**	**89.7**

Table 5: Recall@100 numbers for comparison of ReFIT with both variants of TouR.

3.4.4 Comparison with TouR. We also compare ReFIT with TouR [22], an existing inference-time reranker feedback mechanism. TouR [22] proposes test-time optimization of query representations with two variants: TouR$_{\text{hard}}$ and TouR$_{\text{soft}}$. TouR$_{\text{hard}}$ optimizes the marginal likelihood of a small set of (pseudo) positive contexts. ReFIT shares similarities with TouR$_{\text{soft}}$, which uses the normalized scores of a cross-encoder over the retrieved results as soft labels. Here, we evaluate both approaches on the passage retrieval benchmarks used by Sung et al. [22] – NQ [10] and EntityQuestions [21] – as well as the multidomain BEIR and multilingual Mr.TyDi benchmarks. For NQ and EntityQuestions, we employ the same retriever [9] and reranker [4] as in Sung et al. [22], while for BEIR and Mr.TyDi, we use the models described in §3.1. As shown in Table 5, ReFIT consistently outperforms both TouR variants. We attribute TouR's lower performance to its early stopping criterion for distillation updates; specifically, TouR conducts relevance feedback for three iterations, continuing distillation in each until the top-1 retrieval has the highest reranker score (for TouR$_{\text{soft}}$) or is a pseudo-positive (for TouR$_{\text{hard}}$). In contrast, ReFIT performs more distillation updates within a single iteration (tuned to n=100 updates), making it considerably faster, since each additional iteration of relevance feedback incurs a significant computational overhead.

3.5 Discussion

To better understand how the updated query vector after reranker relevance feedback improves recall, we take a closer look at the query and passage vectors computed for a set of BEIR examples. Figure 3 shows t-SNE plots for four such examples, where each dot represents a vector, and the distance between any two points is their cosine distance. As the figure clearly suggests, the reranker feedback brings the query vector in each case closer to the corresponding

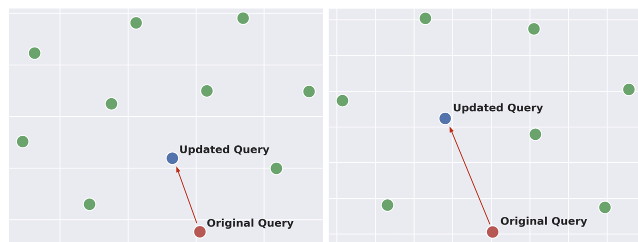

Figure 3: t-SNE plots for examples from BEIR, with the query vectors shown alongside the corresponding positive passages. The updated query vectors after ReFIT are now closer to the positive passages (in green).

positive passage vectors, making the query align with an increased number of relevant passages and consequently improving recall. Across different BEIR datasets, this adjusted query vector is 5–16% closer to the initially retrieved positives. New positives identified by the updated query vector are closest to a passage in the reranker's top 5 in 26% of cases (38% for top 10; 55% for top 20), indicating effective knowledge transfer from the reranker. Across the BEIR datasets, we find that in 24% of the cases where the first-stage retriever retrieves no positive passages, ReFIT can improve recall in a similar fashion. Among all cases where recall improves, however, 75% have at least one positive in the top retrieved results. These results suggest that while the presence of positive candidates in the initial retrieval is useful, reranker relevance feedback approach can also effectively leverage informative negatives to improve the query vector.

4 Conclusion

In this work, we perform an extensive empirical study of ReFIT, our implementation of inference-time reranker relevance feedback, and demonstrate its effectiveness in multi-domain, multilingual and multimodal settings. We show that ReFIT outperforms training-time distillation while introducing considerably low additional latency into the R&R framework. Future work will focus on the potential integration of textual relevance feedback from large language models (LLMs). Another promising direction is improving interpretability by analyzing how relevance feedback affects the importance of individual query terms in the encoded representation.

References

[1] Michael Bendersky, Donald Metzler, and W Bruce Croft. 2011. Parameterized concept weighting in verbose queries. In *Proceedings of the 34th international ACM SIGIR conference on Research and development in Information Retrieval.* 605–614.

[2] Ramraj Chandradevan, Eugene Yang, Mahsa Yarmohammadi, and Eugene Agichtein. 2022. Learning to Enrich Query Representation with Pseudo-Relevance Feedback for Cross-lingual Retrieval. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval.* 1790–1795.

[3] Jonathan H Clark, Eunsol Choi, Michael Collins, Dan Garrette, Tom Kwiatkowski, Vitaly Nikolaev, and Jennimaria Palomaki. 2020. TyDi QA: A Benchmark for Information-Seeking Question Answering in Typologically Diverse Languages. *Transactions of the Association for Computational Linguistics* 8 (2020), 454–470.

[4] Martin Fajcik, Martin Docekal, Karel Ondrej, and Pavel Smrz. 2021. R2-D2: A Modular Baseline for Open-Domain Question Answering. In *Findings of the Association for Computational Linguistics: EMNLP 2021.* Association for Computational Linguistics, Punta Cana, Dominican Republic, 854–870. doi:10.18653/v1/2021.findings-emnlp.73

[5] Sebastian Hofstätter, Sophia Althammer, Michael Schröder, Mete Sertkan, and Allan Hanbury. 2020. Improving efficient neural ranking models with cross-architecture knowledge distillation. *arXiv preprint arXiv:2010.02666* (2020).

[6] Minghao Hu, Yuxing Peng, Zhen Huang, and Dongsheng Li. 2019. Retrieve, Read, Rerank: Towards End-to-End Multi-Document Reading Comprehension. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics.* 2285–2295.

[7] Gautier Izacard and Edouard Grave. 2020. Distilling Knowledge from Reader to Retriever for Question Answering. In *International Conference on Learning Representations.*

[8] Gautier Izacard, Lucas Hosseini, Sebastian Riedel, Piotr Bojanowski, Armand Joulin, and Edouard Grave. 2021. Unsupervised dense information retrieval with contrastive learning. *arXiv preprint arXiv:2112.09118* (2021).

[9] Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense Passage Retrieval for Open-Domain Question Answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP).* 6769–6781.

[10] Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, et al. 2019. Natural questions: a benchmark for question answering research. *Transactions of the Association for Computational Linguistics* 7 (2019), 453–466.

[11] Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. 2022. BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation. In *ICML.*

[12] Shayne Longpre, Yi Lu, and Joachim Daiber. 2021. MKQA: A Linguistically Diverse Benchmark for Multilingual Open Domain Question Answering. *Transactions of the Association for Computational Linguistics* 9 (2021), 1389–1406.

[13] Yuanhua Lv and ChengXiang Zhai. 2009. Adaptive relevance feedback in information retrieval. In *Proceedings of the 18th ACM conference on Information and knowledge management.* 255–264.

[14] Tri Nguyen, Mir Rosenberg, Xia Song, Jianfeng Gao, Saurabh Tiwary, Rangan Majumder, and Li Deng. 2016. MS MARCO: A human generated machine reading comprehension dataset. In *CoCo@ NIPs.*

[15] Rodrigo Nogueira and Kyunghyun Cho. 2019. Passage Re-ranking with BERT. *arXiv preprint arXiv:1901.04085* (2019).

[16] Fabio Petroni, Aleksandra Piktus, Angela Fan, Patrick Lewis, Majid Yazdani, Nicola De Cao, James Thorne, Yacine Jernite, Vladimir Karpukhin, Jean Maillard, et al. 2021. KILT: a Benchmark for Knowledge Intensive Language Tasks. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies.* 2523–2544.

[17] Yingqi Qu, Yuchen Ding, Jing Liu, Kai Liu, Ruiyang Ren, Wayne Xin Zhao, Daxiang Dong, Hua Wu, and Haifeng Wang. 2021. RocketQA: An Optimized Training Approach to Dense Passage Retrieval for Open-Domain Question Answering. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies.* 5835–5847.

[18] Yingqi Qu, Yuchen Ding, Jing Liu, Kai Liu, Ruiyang Ren, Wayne Xin Zhao, Daxiang Dong, Hua Wu, and Haifeng Wang. 2021. RocketQA: An Optimized Training Approach to Dense Passage Retrieval for Open-Domain Question Answering. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Kristina Toutanova, Anna Rumshisky, Luke Zettlemoyer, Dilek Hakkani-Tur, Iz Beltagy, Steven Bethard, Ryan Cotterell, Tanmoy Chakraborty, and Yichao Zhou (Eds.). Association for Computational Linguistics, Online, 5835–5847. doi:10.18653/v1/2021.naacl-main.466

[19] Ruiyang Ren, Yingqi Qu, Jing Liu, Wayne Xin Zhao, QiaoQiao She, Hua Wu, Haifeng Wang, and Ji-Rong Wen. 2021. RocketQAv2: A Joint Training Method for Dense Passage Retrieval and Passage Re-ranking. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih (Eds.). Association for Computational Linguistics, Online and Punta Cana, Dominican Republic, 2825–2835. doi:10.18653/v1/2021.emnlp-main.224

[20] Joseph Rocchio. 1971. Relevance feedback in information retrieval. *The Smart retrieval system-experiments in automatic document processing* (1971), 313–323.

[21] Christopher Sciavolino, Zexuan Zhong, Jinhyuk Lee, and Danqi Chen. 2021. Simple Entity-Centric Questions Challenge Dense Retrievers. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Online and Punta Cana, Dominican Republic, 6138–6148. doi:10.18653/v1/2021.emnlp-main.496

[22] Mujeen Sung, Jungsoo Park, Jaewoo Kang, Danqi Chen, and Jinhyuk Lee. 2023. Optimizing Test-Time Query Representations for Dense Retrieval. In *Findings of the Association for Computational Linguistics: ACL 2023*, Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki (Eds.). Association for Computational Linguistics, Toronto, Canada, 5731–5746. doi:10.18653/v1/2023.findings-acl.354

[23] Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models. In *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 2).* https://openreview.net/forum?id=wCu6T5xFjeJ

[24] Yanmeng Wang, Jun Bai, Ye Wang, Jianfei Zhang, Wenge Rong, Zongcheng Ji, Shaojun Wang, and Jing Xiao. 2021. Enhancing Dual-Encoders with Question and Answer Cross-Embeddings for Answer Retrieval. In *Findings of the Association for Computational Linguistics: EMNLP 2021.* 2306–2315.

[25] Jinxi Xu and W Bruce Croft. 2017. Quary expansion using local and global document analysis. In *Acm sigir forum*, Vol. 51. ACM New York, NY, USA, 168–175.

[26] Jun Xu, Tao Mei, Ting Yao, and Yong Rui. 2016. Msr-vtt: A large video description dataset for bridging video and language. In *Proceedings of the IEEE conference on computer vision and pattern recognition.* 5288–5296.

[27] Nishant Yadav, Nicholas Monath, Manzil Zaheer, Rob Fergus, and Andrew McCallum. 2024. Adaptive Retrieval and Scalable Indexing for k-NN Search with Cross-Encoders. In *The Twelfth International Conference on Learning Representations.*

[28] HongChien Yu, Chenyan Xiong, and Jamie Callan. 2021. Improving Query Representations for Dense Retrieval with Pseudo Relevance Feedback. In *Proceedings of the 30th ACM International Conference on Information & Knowledge Management.* 3592–3596.